Threat-specific engineering · public research

From Threats to Verified Countermeasures

SchoolThreatMitigation | Version 17.0 Revision 1 | Mass-shooting prevention and life-safety assurance

Publication 17.0 Revision 1. Editorial corrections and verifiable advisory reasoning; independent acceptance and site validation remain pending.

A practical specification for translating school mass-shooting hazards into lawful, auditable, independently testable protective requirements.

18
public threat scenarios

15
named control families

5
mandatory approval gates

> **Scope / non-certification.** Illustrative threat scenarios, not a threat forecast or campus vulnerability analysis. The publication includes no site-specific response tactics or sensitive security layouts. No instrument, armed response, canine or AI product is considered qualified merely because this document lists it.

1 · Assurance architecture

Control evidence is evaluated by provenance: (A) enforceable law/code, (B) established prevention or professional guidance, (C) component-level measured performance, (D) credible emerging mechanism, or (E) unvalidated hypothesis. One control may have multiple labels; each claim must specify the relevant endpoint. CISA and USSS NTAC recommend layered, fact-based programs; they do not validate any particular commercial sensor or individual behavioral prediction system.

Public workflow

> Verified concern → protective objective → alternatives including nontechnology → accountable owner → safe synthetic validation → evidence review → limited authorization or stop → monitored reassessment.

2 · Threat scenario index

ID	Threat situation	Control owner / short action	Evidence basis
T01	Household firearm accessible to a minor	Community firearm-safety lead / Voluntary locked and unloaded storage guidance; protect combinations; discounted safe or lock referral	Research-backed prevention mechanism
T02	Firearm in relatives', friends', or shared-care locations	Family engagement coordinator / Universal caregiver discussion guide; voluntary peer-to-peer secure-storage conversation; translated mate	Research-informed extrapolation
T03	Unsecured firearm left in a vehicle	Community retailer and caregiver network / Avoid unattended vehicle storage; voluntary locked storage education; lawful temporary storage referrals	Research-informed prevention
T04	Credible communicated threat or planning leakage	School threat assessment chair / Multidisciplinary, behavior-based assessment; documented corroboration; supportive interventions; emergen	Supported professional practice, not individual prediction
T05	Malicious, mistaken, or uncorroborated accusation	Civil rights officer / independent reviewer / Separate intake from finding; preserve exculpatory information; corrective review, time limits and appeal	Governance / due-process design
T06	Suspected weapon or violence near school arrival/access area	School incident commander / 911 dispatch / Trained human confirmation where safe; dispatch escalation; campus-specific protective action; accessible	Operational guidance; local effectiveness unverified
T07	False or missed acoustic/visual firearm alarm	Technology assurance lead / Independent sensor benchmark; source provenance; human verification; alternate reporting channel	Technology-specific evidence required
T08	Threat reported while groups are in hallways or common areas	Emergency preparedness lead / Preplanned options-based training; intelligible multi-channel guidance; trained staff judgement	General emergency-preparedness evidence
T09	Classroom door hardware fails or impedes egress	Facilities / fire code authority / Code-reviewed listed hardware; scheduled inspection; manual opening and release; accessibility testing	Code/engineering requirement
T10	Person with disability cannot perceive or follow protective instruction	ADA/504 coordinator / Multimodal accessible alerts, individualized support planning, accessible refuge/egress review	Accessibility duty; specific benefit must be tested
T11	Power or communications outage during incident	Facilities / IT / incident command / Tested backup power; manual public address / radio alternatives; fail-safe access and written procedures	Established resilience practice
T12	Spoofed alert, ransomware, or compromised access-control system	CISO / physical security owner / MFA; least privilege; segmentation; signed updates; manual override; offline incident playbook	NIST/CISA control guidance; site testing required
T13	Life-threatening bleeding before EMS arrival	School nurse / EMS liaison / Bleeding-control kits, trained personnel, accessible cache locations, EMS handoff coordination	Established emergency medical preparedness
T14	Delayed police arrival during an active threat	District emergency lead / law enforcement agency / Protective actions, verified alarms, trained	Operational planning; armed response outcome varies

ID	Threat situation	Control owner / short action	Evidence basis
		communications lead, EMS staging; only legally authorized pr	
T15	Canine false indication, injury, or unreliable deployment	Canine program supervisor / independent evaluator / Qualified canine/handler credentialing; validated application-specific reliability; welfare and student s	Limited context-specific evidence; not proven mass-shooting prevention
T16	Misidentified person or unauthorized on-site use of force	Authorized law enforcement / independent oversight / Qualified law-enforcement coordination; identification and restraint standards; independent force review;	Law/policy gate, not demonstrated intervention efficacy
T17	AI-generated accusation, synthetic media, or automated threat score	AI assurance officer / threat team / Source verification, human case review, multiple independent observations and challenge process	Emerging research; model-specific validation needed
T18	After-event reunification confusion or sensitive-data leakage	Reunification coordinator / privacy officer / Single authorized family information lead, secure verification, accessible multilingual updates	Emergency management practice

3 · Scenario-specific control cards

T01 · Household firearm accessible to a minor

Established | ACCOUNTABLE: Community firearm-safety lead

Trigger / verification: Caregiver identifies an unlocked firearm or accessible key/code.

Protective objective: Reduce unauthorized youth access before an incident.

Countermeasure: Voluntary locked and unloaded storage guidance; protect combinations; discounted safe or lock referral.

Independent evaluation: Anonymous opt-in adoption survey, lock voucher accounting, and caregiver comprehension exercise.

Metrics and evidence: Number adopting safeguards; non-identifying participation rate. *Research-backed prevention mechanism.*

Rights/data gate: Do not collect owner identities, addresses, serials or storage inventories.

STOP condition: Ownership reporting required or security intervention based solely on ownership.

T02 · Firearm in relatives', friends', or shared-care locations

Established | ACCOUNTABLE: Family engagement coordinator

Trigger / verification: Caregiver concern about access during visits or custody transitions.

Protective objective: Extend protective practice outside primary home.

Countermeasure: Universal caregiver discussion guide; voluntary peer-to-peer secure-storage conversation; translated materials.

Independent evaluation: Usability test with volunteer caregivers across household types.

Metrics and evidence: Comprehension and voluntary commitment; language accessibility. *Research-informed extrapolation.*

Rights/data gate: Do not infer or disclose custody, family firearm holdings, or guardian status.

STOP condition: Program conditions school access on parental ownership disclosure.

T03 · Unsecured firearm left in a vehicle

Established | ACCOUNTABLE: Community retailer and caregiver network

Trigger / verification: Voluntary report that firearm was previously left accessible in a vehicle.

Protective objective: Reduce opportunistic access and theft exposure.

Countermeasure: Avoid unattended vehicle storage; voluntary locked storage education; lawful temporary storage referrals.

Independent evaluation: Anonymized scenario questionnaire and program distribution audit.

Metrics and evidence: Change in storage intent or reported practice, without individual tracking. *Research-informed prevention.*

Rights/data gate: No vehicle-owner registry or vehicle searches under education program.

STOP condition: Law-enforcement vehicle search derived from education enrollment data.

T04 · Credible communicated threat or planning leakage

Established | ACCOUNTABLE: School threat assessment chair

Trigger / verification: Specific observed statements or behavior with corroborating context.

Protective objective: Review actionable concern early and initiate proportionate supports.

Countermeasure: Multidisciplinary, behavior-based assessment; documented corroboration; supportive interventions; emergency referral only where justified.

Independent evaluation: Tabletop case review with synthetic facts and independent review of proportionality.

Metrics and evidence: Evidence quality, triage timeliness, interventions and overturn rates. *Supported professional practice, not individual prediction.*

Rights/data gate: FERPA purpose limits; protected expression alone is not sufficient.

STOP condition: Unsupported coercion or unsupported record transfer.

T05 · Malicious, mistaken, or uncorroborated accusation

Established | ACCOUNTABLE: Civil rights officer / independent reviewer

Trigger / verification: Anonymous allegation lacking independently corroborated conduct.

Protective objective: Avoid wrongful labeling and retaliatory intervention.

Countermeasure: Separate intake from finding; preserve exculpatory information; corrective review, time limits and appeals.

Independent evaluation: Synthetic false-report exercise with review by legal and student representatives.

Metrics and evidence: False escalation rate, correction timeliness and resolution quality. *Governance / due-process design.*

Rights/data gate: Presumption against disciplinary or police action based solely on allegation.

STOP condition: Student punishment, political profiling or irreversible label without support.

T06 · Suspected weapon or violence near school arrival/access area

Established | ACCOUNTABLE: School incident commander / 911 dispatch

Trigger / verification: Direct human report of potential weapon or danger, not algorithmic inference alone.

Protective objective: Communicate credible urgent risk without exposing additional students.

Countermeasure: Trained human confirmation where safe; dispatch escalation; campus-specific protective action; accessible notification.

Independent evaluation: No-live-weapon notification tabletop; measure decision and alert chain.

Metrics and evidence: p50/p95 notification delay and comprehension; missed recipients. *Operational guidance; local effectiveness unverified.*

Rights/data gate: Avoid publishing campus-sensitive routes or exploitable access points.

STOP condition: Protective action depends entirely on a network sensor or unvalidated classifier.

T07 · False or missed acoustic/visual firearm alarm

Known technology failure class | ACCOUNTABLE: Technology assurance lead

Trigger / verification: Sensor activates without verified incident, or misses simulated signal.

Protective objective: Avoid harmful false response while retaining urgent notification.

Countermeasure: Independent sensor benchmark; source provenance; human verification; alternate reporting channel.

Independent evaluation: Controlled authorized synthetic tests, blinded false-positive and false-negative assessments.

Metrics and evidence: Sensitivity, specificity, alert latency and false alarm workload. *Technology-specific evidence required.*

Rights/data gate: No automatic suspect identity inference or use-of-force trigger.

STOP condition: Vendor cannot disclose validation data; false alerts trigger irreversible action.

T08 · Threat reported while groups are in hallways or common areas

Established | ACCOUNTABLE: Emergency preparedness lead

Trigger / verification: Verified immediate threat with mixed locations and conflicting movement instructions.

Protective objective: Support safer protective action appropriate to each location.

Countermeasure: Preplanned options-based training; intelligible multi-channel guidance; trained staff judgement.

Independent evaluation: Accessible, non-violent exercise with multiple mobility profiles and observers.

Metrics and evidence: Instruction comprehension, conflicts, accessibility and elapsed communication. *General emergency-preparedness evidence.*

Rights/data gate: No publishable room-by-room tactical playbook.

STOP condition: Single inflexible command unnecessarily exposes or traps occupants.

T09 · Classroom door hardware fails or impedes egress

Established secondary safety hazard | ACCOUNTABLE: Facilities / fire code authority

Trigger / verification: Routine inspection finds fault, incompatible accessory, or emergency release failure.

Protective objective: Maintain protective closure while preserving lawful egress and emergency access.

Countermeasure: Code-reviewed listed hardware; scheduled inspection; manual opening and release; accessibility testing.

Independent evaluation: Qualified walk-through under power-off and simulated panic conditions.

Metrics and evidence: Every tested door independently satisfies egress/accessibility criteria. *Code/engineering requirement.*

Rights/data gate: No unsanctioned door lock-in or incapacitating enclosure.

STOP condition: Any tested exit obstruction or inability to release from inside.

T10 · Person with disability cannot perceive or follow protective instruction

Established | ACCOUNTABLE: ADA/504 coordinator

Trigger / verification: Audible-only alert or obstructed accessible exit.

Protective objective: Ensure inclusive, independently usable emergency action.

Countermeasure: Multimodal accessible alerts, individualized support planning, accessible refuge/egress review.

Independent evaluation: Inclusive drill with disabled participants and voluntary accommodations.

Metrics and evidence: Notification reach, support timeliness and documented accessibility defects. *Accessibility duty; specific benefit must be tested.*

Rights/data gate: Do not expose medical or disability details in broad alerts.

STOP condition: Protective protocol excludes or isolates disabled occupants.

T11 · Power or communications outage during incident

Established resilience threat | ACCOUNTABLE: Facilities / IT / incident command

Trigger / verification: Loss of power, WAN, cellular, or security control unit.

Protective objective: Maintain minimum alerts, unlocked legal egress, and command continuity.

Countermeasure: Tested backup power; manual public address / radio alternatives; fail-safe access and written procedures.

Independent evaluation: Authorized loss-of-power functional test with observer and fire authority.

Metrics and evidence: Continuity of essential functions; restoration time and failed transitions. *Established resilience practice.*

Rights/data gate: Backup cannot override life-safety release or logging rules.

STOP condition: Any power loss locks an occupied exit or disables all emergency notification.

T12 · Spoofed alert, ransomware, or compromised access-control system

Emerging and observed cyber threat | ACCOUNTABLE: CISO / physical security owner

Trigger / verification: Unexpected vendor login, unsigned command, inconsistent emergency message.

Protective objective: Prevent cyber compromise from creating unsafe physical behavior.

Countermeasure: MFA; least privilege; segmentation; signed updates; manual override; offline incident playbook.

Independent evaluation: Authorized tabletop and isolated cyber-physical failure injection (no live student impact).

Metrics and evidence: Time to detect/contain; uncompromised exit function; recovery success. *NIST/CISA control guidance; site testing required.*

Rights/data gate: Do not connect surveillance telemetry to general police databases.

STOP condition: Remote operator can close all egress or issue unverified lethal-threat instructions.

T13 · Life-threatening bleeding before EMS arrival

Established | ACCOUNTABLE: School nurse / EMS liaison

Trigger / verification: Observed injury requiring immediate emergency medical response.

Protective objective: Make trained assistance and supplies accessible before professional arrival.

Countermeasure: Bleeding-control kits, trained personnel, accessible cache locations, EMS handoff coordination.

Independent evaluation: Skills-based medical training and equipment inspection; coordinate with qualified clinicians.

Metrics and evidence: Training coverage, kit readiness, performance in safe simulation. *Established emergency medical preparedness.*

Rights/data gate: Do not collect student-specific medical details beyond lawful operational need.

STOP condition: Kits inaccessible or training not sustained.

T14 · Delayed police arrival during an active threat

Established operational constraint | ACCOUNTABLE: District emergency lead / law enforcement agency

Trigger / verification: Dispatch acknowledges incident but units are not yet on scene.

Protective objective: Reduce dependence on a single external response timeline.

Countermeasure: Protective actions, verified alarms, trained communications lead, EMS staging; only legally authorized professional security with oversight.

Independent evaluation: Multi-agency tabletop; timed non-weapon exercise; authority and ID checks.

Metrics and evidence: Reliable dispatch interface, communications robustness, accountable authority. *Operational planning; armed response outcome varies.*

Rights/data gate: No armed deployment outside legal authority, training and explicit policy.

STOP condition: Unqualified staff armed or force authority delegated to software.

T15 · Canine false indication, injury, or unreliable deployment

Specialist capability / uncertain benefit | ACCOUNTABLE: Canine program supervisor / independent evaluator

Trigger / verification: Canine proposed for explosives/firearms screening or a positive indication occurs.

Protective objective: Avoid unvalidated claims and secondary physical harms.

Countermeasure: Qualified canine/handler credentialing; validated application-specific reliability; welfare and student safety review.

Independent evaluation: Double-blind controlled screening trials; documented incident review.

Metrics and evidence: False indication rate, coverage, handler performance, injury near-misses. *Limited context-specific evidence; not proven mass-shooting prevention.*

Rights/data gate: No random intrusive student search based solely on dog alert without lawful grounds.

STOP condition: Insufficient canine validation or foreseeable student bite/exposure risk.

T16 · Misidentified person or unauthorized on-site use of force

High-consequence human/system failure | ACCOUNTABLE: Authorized law enforcement / independent oversight

Trigger / verification: Confusing observations; responder has incomplete incident picture.

Protective objective: Protect innocent people, enforce constitutional limits and prevent friendly-force errors.

Countermeasure: Qualified law-enforcement coordination; identification and restraint standards; independent force review; medical response.

Independent evaluation: Legal policy review, credentials audit and scenario-based decision exercise without weapons.

Metrics and evidence: Misidentification risks, policy compliance, reporting completeness. *Law/policy gate, not demonstrated intervention efficacy.*

Rights/data gate: Never provide target selection, automated force recommendation or unreviewed force authority.

STOP condition: Unlawful force policy; missing chain of command or independent review.

T17 · AI-generated accusation, synthetic media, or automated threat score

Emerging | ACCOUNTABLE: AI assurance officer / threat team

Trigger / verification: Suspicious synthetic audio/video or automated recommendation presents urgent allegation.

Protective objective: Require authentic, corroborated evidence before adverse action.

Countermeasure: Source verification, human case review, multiple independent observations and challenge process.

Independent evaluation: Adversarial synthetic exercise, subgroup error analysis and documentation review.

Metrics and evidence: False escalation, reversals, provenance verification success. *Emerging research; model-specific validation needed.*

Rights/data gate: No student risk scoring, mass surveillance, or AI-only enforcement.

STOP condition: Model recommendation is treated as proof or enables automatic police action.

T18 · After-event reunification confusion or sensitive-data leakage

Established | ACCOUNTABLE: Reunification coordinator / privacy officer

Trigger / verification: Families receive contradictory or over-detailed status messages.

Protective objective: Accountable reunification, privacy and trauma-informed communication.

Countermeasure: Single authorized family information lead, secure verification, accessible multilingual updates.

Independent evaluation: Family-facing tabletop, multilingual review, redacted communication audit.

Metrics and evidence: Family comprehension, rumor correction time, unauthorized releases. *Emergency management practice.*

Rights/data gate: No public posting of pupil identities or sensitive medical data.

STOP condition: Live roster exposed or misinformation uncontrolled.

4 · Engineering requirements and release gates

Control	Intended result	Qualification evidence	Stop condition
P-01 — Voluntary safe storage outreach	Prevent unsupervised youth access	Owner-independent anonymized adoption and comprehension audit	No household ownership registry or mandatory disclosure
P-02 — Temporary secure storage referral network	Offer lawful and voluntary option during acute household risk	Attorney-reviewed transfer/ referral workflow; coverage audit	No physical acceptance of firearms by schools
B-01 — Evidence-based threat assessment	Differentiate credible behavior from stereotypes/rumors	Independent case review; sampled dispositions; appeal test	No adverse finding solely from protected expression or anonymous tip
D-01 — Verified multi-channel emergency alert	Warn and coordinate without automated irreversible action	Blinded alert tests and accessibility review	No deployment with single unverified sensor failure mode
D-02 — Sensor validation and provenance	Understand misses, false positives and adversarial input	Representative independent bench testing	No force or confinement automated by detector
F-01 — Safe classroom protection and egress	Allow protection without trapping occupants	AHJ and disability-access review; outage tests	Stop at any fail-to-egress condition
F-02 — Continuity of critical systems	Keep alerts and releases usable in outages	Power/WAN outage exercise	No failure mode that disables all emergency communications
M-01 — Bleeding-control readiness	Improve immediate medical capability	Training skills test; periodic kit inspection	No inaccessible or unmaintained equipment
R-01 — Professional coordination and lawful force	Reduce confusion and prevent unauthorized coercion	MOU, credentialing, law review, exercise evaluation	No unqualified armed staffing or AI force authority
K-01 — Canine use subject to evidence	Evaluate narrow context-specific canine roles	Independent double-blind tests and welfare checks	No assumed mass-shooting efficacy or dangerous unvalidated search
C-01 — Cyber-physical assurance	Prevent compromised technology from increasing harm	Segmentation, update policy, outage and rollback exercises	Stop when attackers could remotely lock exits
A-01 — AI decision support gate	Keep human responsibility over judgments	Bias, error, uncertainty, adversarial testing	No automatic offender profile, adverse action or force
G-01 — Constitutional and privacy audit	Prevent abuse, retaliation and unlawful transfers	Independent authority, FERPA data-flow, complaint reviews	Any misuse of authority, retaliation or unlawful redisclosure
I-01 — Accessible and inclusive emergency practice	Serve all students and visitors	ADA/504 review and inclusive tabletop drills	Failure to provide meaningful emergency access
L-01 — Shared learning without exposure	Improve with de-identified validated lessons	Evidence/version audit and privacy review	No security-sensitive blueprints or identifying records in public package

5 · Acceptance thresholds and test logic

Set numeric pass criteria **before** testing based on law, AHJ direction, device criticality and measured baseline. Do not retroactively select thresholds from results. Use an independent observer, repeat trials, record failures and quantify uncertainty. Keep disruptive testing off live student operations unless separately approved.

Evidence artifact	Method / metric	Mandatory suspension
Protected-egress sign-off	Fire/building AHJ review; accessible opening and outage trials	Any blocked escape or noncompliant accessibility condition
Alerting and dispatch record	p50/p95 end-to-end delay; recipient coverage; ambiguous alert study	Missing life-critical destinations or unreliable fallback
Sensor performance report	Ground-truth blinded false-alarm/miss evaluation in approved synthetic conditions	Irreversible action from a model score or insufficient independent testing
Safety/rights impact assessment	Purpose specification, complaints, appeal and redress exercise	Identity-based targeting, coerced participation or unlawful redisclosure
Cyber-physical test record	Credential isolation, offline fallback, patch recovery and vendor access review	Remote compromise can block egress or silently modify emergency alert
Personnel qualification and governance	Credentials, MOU, policy, complaint review and case studies	Unauthorized armed deployment, inappropriate use-of-force procedures
Medical readiness log	Skill demonstration, kit inspections, EMS coordination	Known unavailable critical kit or expired readiness checks

Evidence artifact	Method / metric	Mandatory suspension
Privacy-safe after-action report	Redacted aggregate findings, versioned fixes and external review	Student/reporter identity or exploitable campus weakness exposed

6 · Safeguards for specialized response

Canine applications

Separate explosives detection, firearm detection, general patrol, and victim-location concepts; qualification for one does not establish effectiveness for another. Require canine and handler credentials, controlled validation, a documented lawful search basis, injury prevention, supervision and independent incident reporting. Do not describe canine deployment as a proven means of interrupting a school shooting.

Authorized on-site intervention

Decisions about armed security must be taken through separately reviewed statutory authority, qualifications, explicit roles, identification protocols, independent force oversight and integration with dispatch. The public-facing document provides governance and accountability requirements, not engagement, weapons, targeting or entry tactics.

AI and advanced sensors

Confine early experiments to education, maintenance or synthetic decision-support cases. For alerts and individual risk assessment, demonstrate robustness against adversarial and synthetic inputs, human verification, error disparities, auditability and fail-safe operation. No AI-only adverse action, confinement, offender labeling or force decision.

7 · Adversarial assurance without exposing vulnerabilities

Red teams should examine failure classes—spoofing, malicious reporting, outages, social engineering, model hallucination, false alarms and privacy leakage—in isolated environments with written permission. Do not publish operationally exploitable site detail. A second independent team assesses whether the proposed controls themselves create harm.

8 · Source register and scope boundaries

SRC-01 — USSS NTAC, Protecting America's Schools (2019). https://www.secretservice.gov/sites/default/files/2020-04/Protecting_Americas_Schools.pdf. *Historical case analysis and fact-based prevention; not a predictor for individuals.*

SRC-02 — CISA K-12 School Security Guide (3rd ed.). https://www.cisa.gov/resources-tools/resources/k-12-school-security-guide-3rd-edition. *Layered security and campus assessment; does not validate each proposed product.*

SRC-03 — FBI, 2025 Active Shooter Incidents (released 2026-07-30). https://www.fbi.gov/news/press-releases/fbi-releases-2025-active-shooter-incidents-in-the-united-states-report. *2025 total across settings; not a K-12 incident rate.*

SRC-04 — CDC, Preventing Firearm Injury and Death. https://www.cdc.gov/firearm-violence/prevention/. *Public-health guidance for secure storage and child access prevention.*

SRC-05 — Washington RCW 9.41.360. https://app.leg.wa.gov/rcw/default.aspx?cite=9.41.360. *Unsafe-storage liability in specified circumstances; no universal storage method mandate.*

SRC-06 — Washington RCW 28A.320.124. https://app.leg.wa.gov/rcw/default.aspx?cite=28A.320.124. *School safety staff duties, separation from discipline and family complaint processes.*

SRC-07 — Washington RCW 28A.320.1242. https://app.leg.wa.gov/rcw/default.aspx?cite=28A.320.1242. *Annual agreements with public involvement.*

SRC-08 — US Department of Education FERPA threat assessment FAQ. https://studentprivacy.ed.gov/faq/does-ferpa-permit-sharing-education-records-outside-law-enforcement-officials-mental-health. *Purpose limitation and redisclosure conditions.*

SRC-09 — NIST AI Risk Management Framework. https://www.nist.gov/itl/ai-risk-management-framework. *AI governance/validation; not a certification.*

SRC-10 — NIST Cybersecurity Framework 2.0. https://www.nist.gov/cyberframework. *Govern/identify/protect/detect/respond/recover control outcomes.*

SRC-11 — CISA Protecting Our Future: Cybersecurity for K-12. https://www.cisa.gov/protecting-our-future-cybersecurity-k-12. *MFA, known exploited vulnerabilities, backups and IR exercises.*

SRC-12 — U.S. Department of Education Active School Shooter Resources. https://www.ed.gov/teaching-and-administration/safe-learning-environments/school-safety-and-security/preventing-targeted-violence-and-threat-assessment-resources/active-school-shooter. *Behavior-based multidisciplinary TAT and avoiding stereotyping.*

SRC-13 — Washington Department of Commerce Safe Storage Program. https://www.commerce.wa.gov/community-safety/safe-storage/. *Current Washington safe-storage outreach and definitions.*

Prepared 26 October 2026. The 2025 FBI active-shooter total covers multiple site categories, not only schools. Site-specific legal, fire, disability access, engineering and operational approval are not included in this documentation release.